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Ques: Make a documentation or presentation on “Sharding” with the help of some database like MongoDB/NoSQL etc.

Answer: Sharding is a method for distributing a single dataset across multiple databases, which can then be stored on multiple machines. This allows for larger datasets to be split in smaller chunks and stored in multiple data nodes, increasing the total storage capacity of the system.

Sharding is a form of scaling known as horizontal scaling or scale-out, as additional nodes are brought on to share the load. Horizontal scaling allows for near-limitless scalability to handle big data and intense workloads. In contrast, vertical scaling refers to increasing the power of a single machine or

single server through a more powerful CPU, increased RAM, or increased storage capacity.

Sharding in MongoDB

Sharding is a concept in MongoDB, which splits large data sets into small data sets across multiple MongoDB instances. The collection which could be large in size is actually split across multiple collections or Shards. Logically all the shards work as one collection.

When to use MongoDB Sharding?

Sometimes the data within MongoDB will be so huge, that queries against such big data sets can cause a lot of CPU utilization on the server. To tackle this situation, MongoDB has a concept of Sharding, which is basically the splitting of data sets across multiple MongoDB instances.

How to implement Sharding in MongoDB?

Shards are implemented by using clusters which is a group of MongoDB instances.

The components of a Shard include: -

- **A Shard** – This is MongoDB instance which holds the subset of the data. In production environments, all shards need to be part of replica sets.
- **Config server** – This is a MongoDB instance which holds metadata about the cluster, basically information about the various MongoDB instances which will hold the shard data.
- **A Router** – This is a MongoDB instance which basically is responsible to redirecting the commands send by the client to the right servers.

Steps to Set up MongoDB Sharding: -

Step 1: Creating a Directory for Config Server.

For creating a separate directory for config server, we can use **“mkdir /data/configdb”** command.

Step 2: Starting MongoDB Instance in Configuration Mode.

We have to set up server for the configuration, suppose we have a server named “ConfServer” for setting we have use **“mongod -configdb ConfServer: 27019”** command.

Step 3: Starting Mongos Instance.

For starting mongos instance, we can use **“mongos -configdb ConfServer:27019”** command.

Step 4: Connecting to Mongos Instance.

For connecting to Mongos instance, we can use **“mongo-host ConfServer -port 27017”** command.

Step 5: Adding Servers to Clusters.

For adding servers to clusters we can use **`“sh.addShard(“ConfServer:27017”)”`** command.

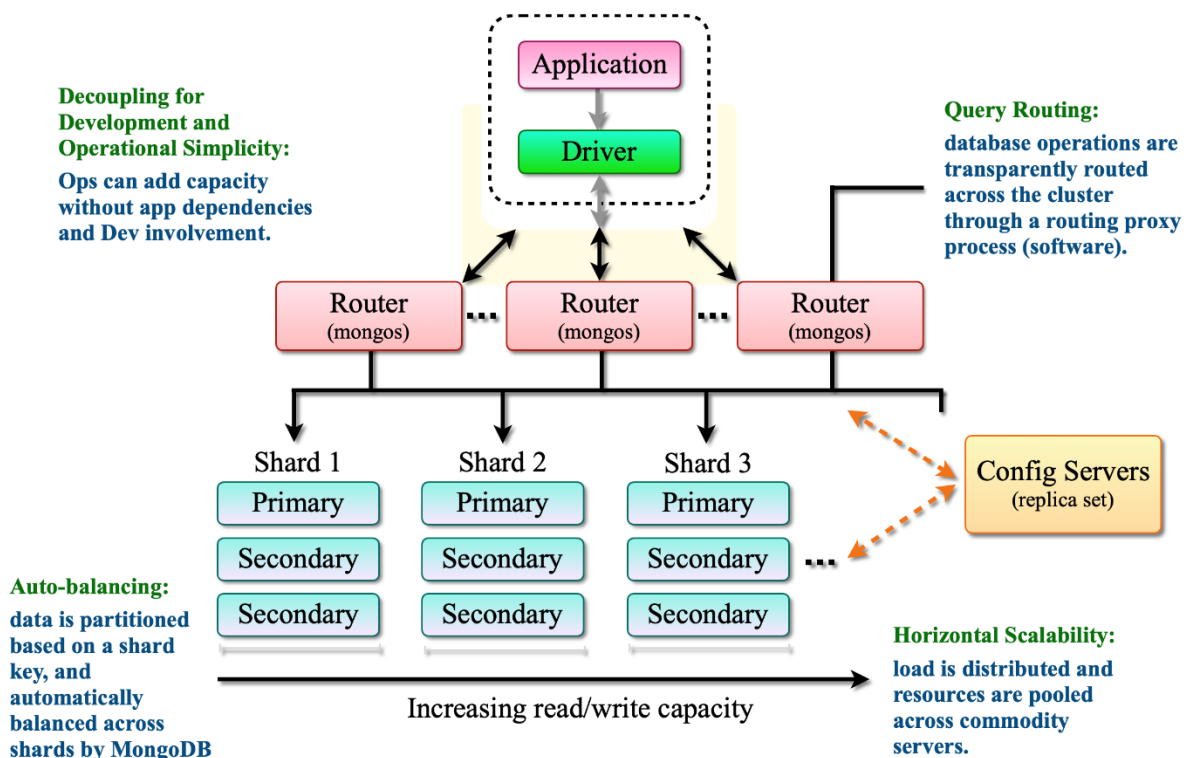
Step 6: Enabling Sharding for Database.

For enabling sharding for database we can use **`“sh.enableSharding(db_test1)”`** command.

You can replace **`“db_test1”`** with your database name.



MongoDB Sharding Architecture



Sharding in NoSQL

Traditional databases (known as relational databases) store interrelated data in a specific format in a single volume. NoSQL is an alternative architecture, wherein data isn't subject to strict formatting, doesn't have to relate to other data, and can be spread across multiple locations. NoSQL databases scale through Sharding, an approach that breaks large data pools into more manageable units.

Working of Sharding in NoSQL

NoSQL databases can perform better than traditional databases in cases of very large volumes of information that only need to be dealt with simply. One example is Twitter, which stores many Tweets but doesn't have to access or manipulate them with complex queries. Additionally, they're much easier to scale, through Sharding, since each shard can be stored on a different database server,

allowing for the dynamic provisioning of additional resources.

Importance of Sharding in NoSQL

Digital transformation means many large enterprises now deal with unprecedented amounts of data, much of it unstructured. NoSQL and Sharding allow enterprises to scale their data repositories and store more types of data while maintaining performance. Apps like Twitter, Tinder, and LinkedIn, multi-user games such as Fortnite, and many others depend on it.

